

Ahasan Habib Rafy

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📍 Dhaka, Bangladesh

Education

Bangladesh University of Engineering and Technology

Dhaka, Bangladesh

B.Sc. in Chemical Engineering

January 2022 – June 2026

CGPA: 3.26/4.00

Relevant Coursework

- To be added

Publications

To be added.

Research Interests

- Water and Wastewater Treatment
- Carbon Capture and Storage
- Materials for Energy and Environment
- Renewable and Sustainable Energy Systems
- Heterogeneous catalysis

Research Experience

Undergraduate Thesis

Title: *Ca-Fe Oxide on Carbon Catalyst for Biodiesel Production*

Supervisor: Dr. Kazi Bayzid Kabir

- Conducted undergraduate thesis research on the development of a Ca-Fe oxide supported on carbon heterogeneous catalyst for biodiesel production.
- Synthesized and evaluated heterogeneous catalysts through laboratory-scale preparation and transesterification experiments.
- Investigated catalyst performance by analyzing biodiesel yield, catalytic activity under varying reaction conditions.
- Performed all experimental work in compliance with laboratory safety regulations and Standard Operating Procedures (SOPs).

Academic Projects

Design Of a 60 MT/DAY Sulfuric Acid Plant

2024

- Designed a sulfuric acid manufacturing plant with a production capacity of **60 MTPD** of 98% sulfuric acid using the Double Contact Double Absorption (DCDA) process.
- Performed process design, material and energy balance calculations, and equipment sizing for catalytic converters, absorbers, heat exchangers, storage tanks, and utility systems.
- Conducted process simulation, process flow development, and preparation of Process and Instrumentation Diagrams (P&IDs) for integrated plant operation.
- Carried out a comprehensive techno-economic feasibility analysis, estimating a total capital investment of approximately **USD 9.99 million** and demonstrating an Internal Rate of Return (IRR) of approximately **19%**.

Additional Projects and Competitions

Deepfake Image Detection using Machine Learning

2023

IEEE Signal Processing Cup

- Developed a deep learning model for deepfake image detection achieving **96%** classification accuracy on the test dataset.
- Performed dataset preprocessing, model training, and performance evaluation using Python-based machine learning frameworks.
- Conducted under the supervision of Prof. Dr. Mohammad Ariful Haque, Department of Electrical and Electronic Engineering, BUET.

Fire Detection and Safety Robot

2019

ESSAB Fire and Safety Robotics Competition – Finalist

- Designed and developed a fire detection and response robot integrating multiple sensors with a microcontroller-based control system.
- Implemented autonomous fire detection logic and hardware control for rapid response applications.
- Achieved **Finalist** (Top 10 out of 134 Projects Nation-wide) position in the ESSAB Fire and Safety Robotics Competition.

Technical Skills

- **Programming:** Python, MATLAB, Java, C#, C, Git, Android Studio
- **Data Science and Machine Learning :** Python Data Analytics Tools(Numpy, Pandas, Seaborn, SciPy, Scikit-learn), Pytorch, Kaggle
- **Simulation Software:** Aspen Hysys
- **Design Software:** AutoCAD, Fusion 360, Blender
- **Documentation:** LaTeX, Microsoft Word
- **Laboratory:** Catalyst Synthesis, Titration, Vacuum Filtration, Calcination, XRD, GC-MS
- **Robotics:** Microcontroller Programming, Sensor Integration

Work Experience

Industrial Trainee

Chattogram, Bangladesh

Eastern Refinery Limited (ERL)

2023

- Completed a one-month industrial training program at Bangladesh's only state-owned petroleum refinery.
- Observed refinery operations including crude oil processing, distillation, and downstream refining processes.

English Proficiency

To be added.